



**BIOCLEAN · SCHOOL & COMMUNITY INNOVATION**

**WASTE → ENERGY**

# **BIOCLEAN SMART BIOGAS SYSTEM**

**(BSBS)**

Combating Air Pollution in Schools and Communities in Enugu State  
Creating a healthy environment for better learning and healthier living.

Presented by

**Chinwuba Chinwendu Jennifer**

Head Girl · Federal Government College, Enugu

## THE PROBLEM

**Air pollution** is the release of substances into the atmosphere in quantities which can be harmful to human health and can cause

### 01 **PM2.5 & PM10**

Tiny dust particles from vehicles and generators; the most dangerous because they reach our lungs.

### 03 **Sulphur Dioxide (SO<sub>2</sub>)**

Released from gas flaring and diesel combustion.

### 05 **Ground-level Ozone (O<sub>3</sub>)**

A major part of smog; formed when sunlight reacts with other pollutants.

### 02 **Carbon Monoxide (CO)**

From the incomplete burning of fuel such as coal, firewood, diesel and petrol.

### 04 **Nitrogen Dioxide (NO<sub>2</sub>)**

A reddish-brown gas from vehicle engines, generators and gas-flaring industries.

### 06 **Methane (CH<sub>4</sub>)**

A colourless, odourless gas from sewage, decomposing waste and gas flaring; forms smog in sunlight.

WHO reports that about **6.7 million premature deaths every year** caused by air pollution.

# Current Situation in My School

Most days in my school — especially within the hostels — my fellow students complain of drowsiness and nausea. This is due to the presence of ammonia ( $\text{NH}_3$ ) and hydrogen sulphide ( $\text{H}_2\text{S}$ ) gas from sewage and open dumps.

## Drowsiness & nausea

Reported especially within the hostels, linked to gases from sewage and open dumps.

## Smoke from burning waste

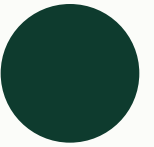
Students feel uncomfortable inhaling smoke when waste is burned on the premises.

## Unbreathable classrooms

Some students no longer enter classes because they cannot breathe comfortably.

**Clean air is not a luxury — it is a learning condition.**

# My Community Challenge



Nowadays, most homes in my community utilise charcoal, stoves and generators for cooking and powering their houses. As we all know, coal burns in a limited supply of air to release carbon monoxide (CO), a colourless and odourless gas which, when inhaled excessively, accumulates in the body and reacts with the haemoglobin of our blood to form carboxyhaemoglobin. This cuts down the level of oxygen delivered in the body, leading to severe cardiovascular issues or even death.

## THE COMMUNITY NEED

**Safer waste handling + cleaner energy + better sanitation → healthier, more productive homes and schools.**



## 01 Health effects

Air pollution affects our health through diseases like asthma, pneumonia and lung cancer. Families are exposed to serious, long-term health risks.

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## 02 Academic effects

Students find it difficult to learn in polluted environments, leading to poor grades by the end of the term or session. Teachers may also lose interest in teaching, causing poor educational outcomes.

## 03 Economic effects

Families spend their income on treatment and medical expenses, reducing money available for other needs. Workers also lose productive days to sickness.

## 04 Social effects

Air pollution reduces interaction among people, who fear stepping out to inhale smoke, smog or bad odour. Students face embarrassment and stigma over fears of airborne disease.

# The Research and Science Behind the Innovation: Bioclean Smart Biogas System (BSBS)

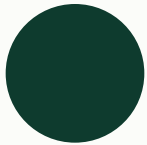
## What is biogas?

Biogas is a renewable fuel produced when microorganisms break down organic waste without oxygen.

## The process

| 5 Steps        |                                                                                                    |                                                                     |
|----------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| C5 Projects    |                                                                                                    |                                                                     |
|                | Anaerobic → Energy                                                                                 |                                                                     |
| Inlet plumbing | Waste collection system from toilets, kitchens and livestock farms; a mobile tank and farm hopper. | Plumbing from hostels, classrooms and refectory kitchens.           |
| Pre-treatment  | Pre-filter and mixing tank.                                                                        | Removes plastics and sand; mixes waste with water for the microbes. |
| Process        | Modular digestion, accelerated microbe action and temperature control.                             | 7-day gas production instead of 30 days.                            |

# Project Cost and Funding Plan



|                                                                           | Costs             | Stages                                   |
|---------------------------------------------------------------------------|-------------------|------------------------------------------|
| Equipments And Materials                                                  | ₦1,870,000        | 70%.                                     |
| Installation and labour — Engineers, Plumbing, Safety testing, Labourers. | ₦580,000          | 21%                                      |
| Contingency and setup permits, Microbes, App monitoring.                  | ₦230,000          | 7-day gas production instead of 30 days. |
| <b>Total CAPEX</b>                                                        | <b>₦2,650,000</b> | <b>100%</b>                              |

|                                             | Costs             | Stages |
|---------------------------------------------|-------------------|--------|
| Winning the National Head Girl Prize        | ₦1,000,000        |        |
| School Management and P.T.A + CSR Partners. | ₦1,650,000        |        |
| <b>Total</b>                                | <b>₦2,650,000</b> |        |

THE BUSINESS CASE

# Financial Viability and Sustainability

Annual net benefit to school

**#1,880,000**

**#1,860,000 Savings**

No LPG and no waste-truck fees.

**#320,000 Revenue**

Organic fertilisers and carbon credits.

**#80,000 Running cost**

Maintenance and microbes.

**#1,880,000 Net gain every year**

**17**  
Reported annual net benefit.  
Months payback

## Key metrics

As the Head Girl of my college, I will sustain this innovation by:

- 01** Organise 10 reliable students from our Green Club to manage its operations. generates savings for 15 years.
- 02** Train two (2) juniors for continuity.
- 03** Enhance scalability so more models can be replicated in other schools and communities.

## THE SOLUTION

# How My Smart Bioclean Biogas System (BSBS) Combats Air

### BEFORE

#### Health

Biogas burns clean and releases no smoke into the atmosphere, reducing health risks from firewood, charcoal and generator fumes.

#### Agricultural

The residue left behind after production is collected by the slurry outlet and dried on the drying bed before it is supplied to farmers. This encourages farmers to cultivate more for bountiful harvests, fostering agriculture.

#### Environmental

BSBS traps methane, a greenhouse gas and major contributor to climate change. Moreover, it provides a clean, renewable alternative to firewood and charcoal.

### AFTER

#### Economic

Biogas comes from decomposing organic materials, so My BSBS is affordable for families and schools. A 5m<sup>3</sup> model in each home brings a net benefit of ₦1,880,000. It would minimise the use of petrol and cooking gas, and create employment for skilled residents.

#### Social

Students no longer feel isolated or stigmatised, because our environment provides everyone with clean air. People need not stay indoors because of polluted air — with My BSBS in every home, air pollution decreases.

THANK YOU

# Conclusion and Gratitude

## 01 The problem is real

300 students suffer daily from gas fumes; waste causes bad odour and ₦1.56m is wasted every year.

## 02 The solution works

Biogas turns our biggest problem into our biggest asset: clean cooking, clean hostels, plus ₦1.8m net gain every year.

## 03 The plan is ready

₦2.65m total cost: ₦1m from WILAN, ₦1.65m from partners; 17-month payback and 15 years of impact.

## 04 The leadership is here

With the BSBS Green Club and annual training, this impact will outlive me and inspire other schools in Enugu.

**Invest ₦1,000,000 in me today. In return, WILAN gets a national model of student-led climate action that saves money, saves the planet and saves lives.**

This is not just a project — this is my legacy.

**Let's Breathe In Clean Air together.**

Thank you, WILAN, for this opportunity to change my school, community, nation, continent and the whole world.

**BIOCLEAR SMART BIOGAS SYSTEM · BSBS**